WORKING SCIENTIFICALLY

I Am a Scientist

Looking closely, asking questions, predicting, fair tests, measuring and recording


magnifying glass


test tube


ruler


thermometer

In this pack

Learn it: I am a scientist! ★☆☆
Fact check: Science tools ★☆☆
Think it through: Is it fair? ★★☆
Investigate: Which ball bounces highest? ★★☆
Real-world case study: Rumi the bird watcher ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · W1 · WORKING SCIENTIFICALLY

I am a scientist!

★☆☆ Grade 2

Name: Class: Date:

Learning goals

- Use my senses to look closely and ask questions.
- Make a prediction and test it.
- Measure, record and tell others what I found.

Scientists are people who are curious about the world. They look closely, ask questions and try things out to find answers. You can be a scientist too!


look closely


ask a question


try it out


measure


record


tell others

1. Look closely

Use your **senses**: look, listen, touch and smell (but never taste in science!). A **hand lens** makes small things look bigger.

2. Ask a question

Good questions start with **What**, **Why**, **How** or **Which**. "Which ball bounces highest?" "Does a plant need water to grow?"

3. Predict

A **prediction** is what you think will happen. "I think the big ball will bounce highest."

4. Try it out fairly

Change only **one** thing. Keep everything else the **same**. If you test balls, drop them from the same height on the same floor.

5. Measure and record

Use simple tools: a **ruler** for length, a **thermometer** for how hot or cold, a **timer** for time and **scales** for how heavy. Write or draw what you find in a **table** or a **chart**.


ruler


thermometer


timer


scales

6. Tell others

Share what you found with drawings, charts and words. Did your prediction come true?

> ⚠ **Stay safe**
>
> Never taste or smell things in science unless your teacher says it is safe. Wash your hands after touching soil or animals.

> 🕮 **Key words**
>
> **observe** look closely
> **predict** say what you think will happen
> **measure** find out how long, heavy, hot or how much
> **record** write or draw what you found
> **fair test** change only one thing

FACT CHECK · W1 · WORKING SCIENTIFICALLY

Science tools

★☆☆ Grade 2

Name: Class: Date:

Learning goals

- Name science tools.
- Match tools to jobs.

1 Name the tool. [5]







2 Draw a line to match the tool to its job. [5]

Draw a line to match each pair, or write the letter.

ruler ●	● **A** how much time
thermometer ●	● **B** how heavy
timer ●	● **C** see small things
scales ●	● **D** how hot or cold
hand lens ●	● **E** how long

3 Which sense do you use? Write see, hear, touch or smell. [4]

I want to find out...	Sense
the colour of a flower	
if a stone is rough	
which bird is singing	
if the mango is ripe	

THINK IT THROUGH · W1 · WORKING SCIENTIFICALLY

Is it fair?

★★☆
Grade 2

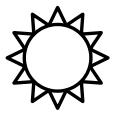
Name: Class: Date:

Learning goals

- Spot a fair test.
- Make predictions.

1 **Rina and Sami race toy cars down a slope. Rina pushes her car. Sami lets his car go. Is it fair? Why?** [2]

2 **Predict: Which will dry faster, a wet towel in the sun or a wet towel in the shade?** [1]

 in the sun	 in the shade

3 **Draw what you would see if you looked at your thumb through a hand lens.** [2]

My thumb

4 **Write one question you want to find out about animals.** [1]

INVESTIGATE · W1 · WORKING SCIENTIFICALLY

Which ball bounces highest?

★★☆ Grade 2

Name: Class: Date:

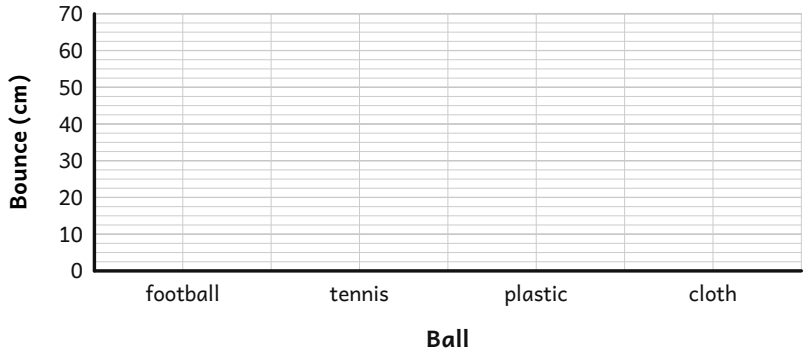
Learning goals

- Do a fair test.
- Make a bar chart.

Class 2A dropped four balls from the same height onto the floor and measured the bounce with a metre ruler.

1 **Here are the results. Draw the bar chart.** [3]

Ball	football	tennis ball	plastic ball	cloth ball
Bounce (cm)	60	50	30	10



2 **Which ball bounced the highest?** [1]

.....

3 **Which ball bounced the lowest?** [1]

.....

4 **What did they keep the same?** [1]

- ◯ the height they dropped the balls from
- ◯ the type of ball
- ◯ the colour of the balls

REAL-WORLD CASE STUDY · W1 · WORKING SCIENTIFICALLY

Rumi the bird watcher

★★★ Grade 2

Name: **Class:** **Date:**

Learning goals

- Read a true-to-life story.
- Count and record.
- Ask questions.

Science · Maths · Nature

Rumi lives in a village near a big lake. Every morning before school she sits quietly by the window with her grandfather's old notebook. She looks at the birds in the mango tree and counts them.

Rumi draws each bird and writes the date. She sees that in winter there are many ducks on the lake, but in summer they are gone. "Where do they go?" she asks. Her teacher says, "You are thinking like a scientist!"

A story about observing nature. Numbers below are example data.

1 **Rumi counted birds for five mornings. Make a tally and write the totals.** [4]

Bird	Tally	Total
crow	IIII I	
myna	IIII	
magpie robin (doyel)	II	
kingfisher	I	

2 **Which bird did Rumi see most?** [1]

.....

3 **Which science skills did Rumi use? Tick them.** [3]

Skill	Tick
looking closely	
counting	
asking a question	
tasting	

Show what you know

★★★
Grade 2

Name: Class: Date:

Learning goals

- Use science words.
- Answer questions.

1 Find the words.

F	P	N	T	T	E	S	T
T	J	T	R	I	A	C	L
R	R	F	C	G	M	B	S
S	G	L	R	L	P	E	O
E	U	O	A	U	B	H	R
N	E	O	P	O	L	T	R
S	S	K	L	L	N	E	F
E	S	B	F	O	A	D	R

guess look ruler sense test timer

2 Which tool measures how hot something is? [1]

◯ ruler ◯ thermometer

◯ scales

3 A fair test changes... [1]

◯ one thing ◯ everything

◯ nothing

4 Challenge. Draw a table to record how many legs these animals have: spider, bird, cow, ant. [2]

My table

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
use my senses to look closely	○	○	○
ask a science question	○	○	○
make a prediction	○	○	○
do a fair test	○	○	○
measure with a tool	○	○	○
make a bar chart	○	○	○